LogicGrid
SYSTEMS FOR LOCAL GOVERNMENT

SYSTEM OVERVIEW & FEATURE REFERENCE

eLGU Citizen Portal

A two-tier electronic government system for a Philippine municipality — and for every barangay inside it.

One codebase serves five portals: a mobile-first app for residents, a console for each municipal office, a console for each barangay, a read-only view for the barangay council, and a public document check that needs no account at all. This document is the complete reference — what the system is, who may do what, and what every control on every screen does, shown with the icon it actually carries.

6 municipal offices	**17** municipal services	**6** barangay certificate types	**8** sectoral & poverty programmes
5 portals from one codebase	**165** controls, documented here	**50** database tables	**1,639** automated assertions

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Sample configuration: Malasiqui, Pangasinan

What this is, and what it is built around

A working system, not a prototype: a Node and PostgreSQL API with 50 tables, a React progressive web app, real PDF documents with scannable verification, and 1,639 automated assertions run against a live database.

Both tiers of government

Most LGU portals stop at the municipal hall. This one also holds the barangay — the clearance, the certificate of residency, the record of inhabitants, the blotter, the board of notices. That is the tier most residents actually deal with in a given year, and the two constantly reference each other: the municipal business permit asks for the barangay clearance first.

Built around the wasted trip

The expensive part of dealing with an LGU is not the fee. It is turning up at the wrong counter, or at the right one without the document that had to come first. Every service carries an ordered, bilingual route; ready dates are computed over the municipality's own holiday calendar; and the clock stops while a request waits on the resident to pay.

The server establishes its own facts

The browser never asserts a fee, a workflow position, a municipality or a payment. A fee is read off the request's own assessment; a payment settles only on a signed callback from the gateway; the barangay is stamped from the register. Roles are a server-side fact — the shell a user gets is a rendering convenience, never a control.

It works on one bar of signal

The emergency hotlines, the digital ID and service search all read from a device cache. A half-filled form is written to the phone on every keystroke, and a submission made offline queues and sends itself when the signal returns, carrying the time it was actually filled in.

PhilSys has been removed

The number was typed at registration and stored, and nothing in this system ever called a PhilSys API — while the ID card printed "PhilSys Verified" over it, the profile showed a green tick, and the counter check returned its last four digits. A badge asserting an identity check nobody performed is worse than no badge, because a clerk reads it as one that happened. The claim, the database column and the masked digits are all gone. What remains is the verification this app really does perform: the ID's own code, which lives ninety seconds, is consumed on first use and is checked server-side.

The five portals	Controls
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1 Resident app

Who uses it. Any resident of the municipality. They register themselves.

A mobile-first progressive web app. Five items in the bottom bar; everything else is reached from them.

The bottom bar

Five destinations, always on screen. Both tiers of government sit under one Services item — a second one would be a second place to make the same choice.

Home
The dashboard: digital ID, the emergency button, the two tier buttons, recent requests.

Services
The catalogue — municipal offices and barangay certificates, with search.

Requests
Everything filed, plus drafts not yet submitted.

Appointments
Booked visits and their queue numbers.

Profile
The ID, complaints, data rights, help, language and text size.

Home

The emergency button is the only red control on the screen, and the two tier buttons are the largest. Nothing else competes with them.

Notifications
Opens the notification list, with an unread count on the bell.

Emergency hotlines
Opens the numbers. It never dials by itself — an app that places a call on a tap is one that rings the fire brigade from a pocket.

Municipal
17 services across the hall's 6 offices.

Barangay
The resident's own barangay: notices first, then certificates.

Digital ID card
Tap the card to open the full ID.

Install this app
A dismissible banner, shown only where the browser says installing is possible.

Services and the route to a document

Search runs on the device against the cached catalogue, so it is instant and works on one bar of signal.

- **Search**
  Matches what residents actually type — "sari-sari store", "amilyar", "ayuda" — in both languages.
- **Clear search**
  Restores the office tiles.
- **Tier switch — municipal**
  Changes tier without walking back to the dashboard.
- **Tier switch — barangay**
  The same pair as the dashboard, so the tier you are in is always on screen.
- **How to get this**
  Opens the ordered, bilingual route to the document.
- **A step that is itself a document**
  Becomes the next tap, opening that document's own route. Chains go four deep.
- **Ring the office**
  The guide ends with the number, not a disclaimer: "confirm before you travel" is useless without it.

Filing a request

Fields are generated from the catalogue, so a new service needs no new screen.

- **Requirements checklist**
  What to bring, shown before the form.
- **If you apply today**
  A working-day estimate over the municipality's own holiday calendar.
- **Saved**
  The draft reached the server.
- **Saved on this device**
  Written to the phone on every keystroke; the server copy follows, debounced.
- **Discard draft**
  Throws away the resumed draft.
- **Submit**
  Validated in the browser and again on the server.
- **Save and send when online**
  Offline, it queues and appears in the list at once; the server later records when it was actually filled in.
- **Attach the files now**
  Offered on success where the service has requirements — the office cannot act until they are in.

A request, open

Every workflow step says whose move it is in words, never by colour alone.

- **Take photo**
  Hands the job to the phone’s own camera, which focuses, uses the flash and applies the phone’s processing.
- **Attach file**
  A PDF or picture already on the device. 10 MB each, 10 per request, type read from the bytes.
- **Remove attachment**
  Takes a file back off the request.
- **Pay now**
  Opens payment once the office has assessed the fee.
- **Download document**
  The released PDF, with seal, signatures and a verification QR.
- **Download receipt**
  The electronic official receipt for a settled payment.
- **Book a visit**
  An appointment about this request, rather than starting again.
- **Rate this service**
  Asked once, when the office has finished — including on a rejection.
- **Continue this draft**
  Reopens a half-filled form from the Drafts tab.

Paying

Nothing the browser sends moves a peso. The server prices the intent and only a signed callback settles it.

- **Choose a provider**
  Hands off to the gateway.
- **Confirm and pay**
  Creates an intent for an amount the server read off the assessment.
- **Waiting for the gateway**
  Polls the server rather than trusting its own return — close the tab and you are still paid up.
- **Get the receipt**
  Written in the same transaction that settled the payment.

The barangay hub

The notice board sits above the certificates: what the barangay is telling you comes before what you can ask it for.

- **Notices**
  The board. Urgent ones carry a word and an icon in amber — never colour alone.
- **Request a certificate**
  One purpose field, because that is what a barangay certificate actually asks.
- **Valid until**
  Stamped when the barangay marks it ready. A lapsed one says so in words and asks you to request a fresh one.
- **The hall**
  Address, hours and phone on the "ready" row and in the notification, so you know before you travel.
- **Download certificate**
  Available once released — the barangay takes its fee at the hall on collection.

Appointments

A day the hall is shut cannot be booked; capacity is enforced under a lock.

- **Book an appointment**
  Office, service, day, time — optionally about a request the office already holds.
- **Pick a day**
  Weekends and declared holidays are closed, read from the same calendar as every ready date.
- **Pick a time**
  A full or past slot is greyed, not hidden, so you can see that 8:30 has gone.
- **Confirm**
  Returns a queue number drawn from a counter, so a cancellation frees the slot but not the number.
- **Where to go**
  The office this booking is with.
- **Cancel**
  Asks once, then frees the slot for somebody else.

The digital ID

There is no national-ID integration and the app no longer implies one. What proves this card is its own code.

The card face
Name, address, date of birth and the entitlements live on the record right now. A lapsed one is dropped, so it cannot read as current at a counter.

Show verification code
Lives 90 seconds and is consumed the first time a clerk checks it. A screenshot is stale before it reaches a window.

New code
Issuing one retires any still outstanding, so a code left open in another tab is not a second key.

Take a selfie / choose a photo
Cropped square and scaled before upload. Only the resident can ever read it back.

Remove photo
Takes the portrait off the ID.

Emergency numbers on the ID
Cached on the device, because somebody needing one is very often somewhere with no signal.

Emergency and reporting

Reporting sits under the hotlines, never above. A resident watching a fire should be dialling.

Call a hotline
Municipal numbers first, nationwide below, so you can see what still answers if the municipal line is down.

Report an incident
A flooded road, a downed line, uncollected rubbish. The screen says twice that nobody is dispatched by this.

Take photo
Evidence from the phone’s own camera.

Record video
Streamed to disk rather than held in memory, so one upload cannot take the server down.

Send report
Returns a reference. Reports land with the administrator, because an incident is not any one office’s work.

Profile and the rights screens

Reached from Profile rather than the bottom bar — these are screens you go looking for, not ones you pass through.

- **Digital ID & verification code**
  Opens the full ID.
- **Complaints**
  File one against a named office under RA 11032, with a visible deadline in working days.
- **My data**
  Export, correct and close — the three RA 10173 rights as buttons.
- **Privacy notice**
  What is held, and who the Data Protection Officer is.
- **Help & support**
  Office address, hours, phone and email.
- **English / Filipino**
  Switches the whole app, including dates and durations.
- **Text size**
  Normal, Large, Largest. Each option renders at the size it selects.
- **Push notifications**
  Only rendered where push is actually configured — a toggle that cannot work is worse than none.
- **Install the app**
  Never dismissed here, and confirms "Installed" once it is.
- **Sign out**
  Clears the session on this device.

My data — the three rights

Erasure is the right the app cannot grant in full, and it says so first.

- **Export my data**
  One JSON file: profile, every request and its history, attachments listed, drafts, appointments, complaints, feedback, notifications, programme records, ID checks and devices.
- **Correct my details**
  Address and phone — facts the resident knows better. A name goes through the Civil Registrar, and the screen says so rather than leaving a dead field.
- **See what closing would do**
  Names what would be deleted and what would be kept, with the reason, before anything happens.
- **Close my account**
  Closes the sign-in, erases contact details and the portrait, cancels upcoming appointments and releases the email address.

2 Municipal staff console

Who uses it. Issued accounts for the hall’s 6 offices. Three tabs are the administrator’s alone.

A desk-width console. Which tabs render depends on the account — and the API refuses the rest regardless. The hidden tab is the courtesy; the refusal is the control.

The tabs

Queue, Today, Verify ID, Complaints and Reports are every officer’s. Payments is the treasury’s. Incidents, Accounts and Municipality are the administrator’s.

Queue
This office’s requests, filtered by status. Scoped to the departments the account serves — naming another is a refusal, not an empty list.

Today
The counter’s day: one office, in time order, queue number, who is expected and what for — and nothing else about them.

Verify ID
Check a resident’s ID code. Opens pre-filled and runs itself when the clerk arrived by scanning.

Complaints
Oldest-open-first. Anything past its deadline says "Overdue" in words, not only in colour.

Reports
What this office did, and whether it was on time.

Payments
Treasury and administrator only. Reconciliation, with the gateway reference on every line.

Incidents
Administrator only. Open-first then oldest-first, so the list reads as the order to work it.

Accounts
Administrator only. Issue, deactivate, reactivate and re-invite staff.

Municipality
Administrator only. The hall’s own data, in seven panels.

Working a request

Workflow progression belongs to the server. The console never sets the step itself.

Assess
Sets the fee. This figure, not the browser's, is what the payment is later priced from.

Advance
Moves the request one step and notifies the resident in the same transaction.

Reject
Requires a reason the resident can read.

Release
The final advance. Renders and stores the PDF with the seal, both signature blocks and the verification QR.

Download attachment
Served as a download only, never inline.

History
Every move, who made it and when.

Programme records
Grant or revoke a sectoral or poverty record — CSWDO and admin only, and reached through a request so no endpoint lists residents.

Withdraw this document
Final, and needs a reason. Only the office that released it; another office gets "not found".

Reissue
Mints a new document with a new code, printed "REISSUED COPY". Reachable only from a withdrawal, and only while nothing has replaced it.

Today, Verify ID and Complaints

What happened at a window should not be quietly rewritten later in the day, so the closing actions are final.

- **Arrived**
  The resident has presented themselves.
- **Served**
  Closes the visit. Final.
- **No show**
  Closes the visit. Final.
- **Check this code**
  Returns a face, a name, where they live and the live entitlements — no email, no phone, no request history.
- **Unknown / expired / used**
  Three answers, not one failure, because they mean different things at a counter.
- **Reply to a complaint**
  Accepted once. The server refuses a second — an office rewriting what it told a resident is what the mechanism exists to prevent.

Reports and Payments

Two turnaround clocks, because either alone lies. The gap between them is itself the finding.

- **Office days**
  Median and p90 while the office actually held the request.
- **Resident days**
  Submission to release regardless. A wide gap is a payment step residents struggle with, not a slow office.
- **On time**
  Judged against the catalogue’s own maximum — the same promise the resident was quoted.
- **Download CSV**
  The same figures, for a spreadsheet.
- **Settled in this period**
  Leads with the total; every line carries the gateway reference.
- **Needs a person**
  Anything that did not end cleanly, deliberately not limited to the period — a payment stuck since last week is what reconciliation exists to surface.
- **Official receipt**
  Opens the eOR for a settled line.
- **Fee changes**
  A read of the audit trail, with the old figure beside the new one.

Municipality — the seven panels

Every write here is audited in the same transaction as the change, with the row as it was and the actor's name beside their id.

Emergency hotlines
Add, edit, reorder, remove. The panel asks whoever is editing to ring the number before saving.

Holiday calendar
Declaring one moves dates already promised — intended, because the hall really is shut.

Office details
Address, hours, phone, email, and who signs for that office — plus the hall's own signatory above them.

Official receipt
What prints on an eOR: address, TIN, treasurer, series prefix, seal and signature. The municipality's name is not editable — that belongs to the register.

Barangay certificates
The shared catalogue every barangay issues, including the attestation text, whose placeholders are checked at the save rather than at the print.

Document checks
How often each document has been verified. A prompt to look, never a finding — a busy permit and a copied one are identical from here.

What changed
The audit trail, showing only the fields that actually moved.

Upload seal / signature
PNG and JPEG only by the leading bytes, virus-scanned, 2 MB — and a PNG must actually decode, because one damaged file used to take the server down at release.

3 Barangay console

Who uses it. The barangay's own administrator (captain or secretary) and its clerk.

A separate shell, not a tab in the municipal one. A barangay secretary does not work a municipal queue and has no business seeing one — separate shells mean there is no filter to get wrong.

The tabs

The first five are both roles'. The last three are the administrator's: what the barangay charges, who works there, and what it has done.

Requests
The certificate queue: pending, processing, ready, released, rejected.

Notices
The board of notices residents see in their barangay hub.

Blotter
Katarungang Pambarangay: the book a dispute is written in, and the summons that call both sides.

Households
The Record of Barangay Inhabitants — who lives where, and the counts the barangay plans from.

The hall
Captain, secretary, address, hours, phone, seal and the two signatures. Both roles read; only the administrator writes.

Summary
Administrator only. What the office issued and collected.

Certificates
Administrator only. Override the fee on a shared type; add, edit and withdraw the barangay's own.

Officers
Administrator only. Issue the barangay's own clerks and kagawads — but never another administrator.

Moving a certificate

The order is enforced on the server, not merely rendered: "ready" means come and collect it, and a resident told that too early makes a wasted trip.

Processing
The secretary has taken the certificate up.

Ready
Prepared and dated. This is where the validity period is stamped — so a resident who collects three weeks later holds a certificate three weeks into its life, exactly as the paper says.

Released
Handed over, fee taken at the hall. This is where the PDF is written, with the seal and both signatures.

Reject
Requires a reason the resident can read.

Download certificate
The stored PDF — frozen at issue, so an election never reprints it over somebody else's name.

Withdraw
Administrator only. Both roles hand a certificate over; taking the barangay's word back is a decision.

Reissue
Administrator only. Carries the same valid-until as the one it replaces — correcting a mistake is not renewing.

The board of notices

Every other barangay feature refuses to be readable; this one inverts all of it. A notice is worth nothing until the people it concerns have seen it.

Post a notice
Title, body, optional expiry. Appears in every resident's barangay hub.

Mark urgent
Administrator only. Writes a notification for every resident of the barangay, in the same transaction as the notice.

Reached N residents
The confirmation waits to be dismissed — the one screen stating the blast radius should not vanish on a timer.

Edit
Allowed, and the board then says "edited". Urgency is not editable: the notification went or it did not.

Take down
Administrator only. Keeps the row in the archive — what the barangay said last month is a thing somebody asks about.

Blotter and summons

The most defamatory data in the system. No resident route reads it, it is not in the privacy export, and municipal staff get refused — administrator included.

- **Record a matter**
  Complainant, respondent, contacts, nature, narrative — all free text, never linked to accounts.
- **Search the book**
  Searching names, bounded to the barangay doing it.
- **Issue a summons**
  Administrator only — it goes out over the Punong Barangay's name and compels two people to appear.
- **Send the reminder**
  The only thing here that notifies anybody. Who is asking, when, where, the reference — never the narrative and never the other party's name.
- **Record service**
  Kept apart from the outcome: a respondent who never received the summons did not fail to appear.
- **Record what happened**
  Administrator only. One pending calling at a time.
- **Close the matter**
  Final, cancels any outstanding hearing, and must say how it ended.

The register of inhabitants

The one place in the app that is deliberately a directory. Its containment is the barangay boundary — nothing here is readable by the municipality, by another barangay, or by any resident.

- **The figures**
  Population, households, age bands, per purok — computed over the active register on every read.
- **Without a date of birth**
  Reported rather than hidden: counted in the population and in no age band. A feeding programme planned off a silently short count is the harm.
- **Add a household**
  The head becomes a member automatically — a register whose head is only a column cannot count its own population.
- **Add a member**
  Free text, never linked to an account. A link would hand a clerk a view onto the user table.
- **Edit a member**
  PWD and 4Ps ticks are labelled "(noted)" so a checkbox can never read as an entitlement at a counter.
- **Take out of the register**
  Administrator only, and needs a reason. Nothing is deleted — next year's figures have to be comparable with this year's.

The hall, and the barangay’s own accounts

A barangay issues its own clerks and councillors, because the hall does not know who holds a barangay office this term and the captain does.

Correct the hall’s details
Captain, secretary, address, hours, phone, email. The barangay’s own name is not editable — that belongs to the register.

Upload seal and signatures
The Punong Barangay’s and the Secretary’s. There is no third slot.

Issue an officer or kagawad
Starts with an unusable password; the holder chooses their own from an emailed link, and the screen says whether it sent.

Revoke
Spends any outstanding invitation, because a live reset link is a way back in. An administrator’s row is "not found" here — which is what stops a captain and a secretary revoking each other.

4 Council view

Who uses it. A member of the Sangguniang Barangay.

Its own mount with its own gate, not a reduced clerk. Putting a third role on the clerk's gate and carving it back out route by route would leave the blotter one forgotten check away from a councillor.

What a kagawad sees

The figures come from the same code as the secretary's own screen, so the council and the clerk never quote different population numbers at each other.

- **Summary**
  What the office issued and collected.
- **Register figures**
  Population, age bands, households per purok — counts only.
- **The notice board**
  Exactly as residents see it.

What is withheld, and why

Read-only is deliberate and conservative in the right direction: widening this later is one route; narrowing it after seven people have been using it is not.

- **No certificate queue**
  Individual residents and what they asked for.
- **No household names**
  The register as people rather than as counts.
- **No blotter, entirely**
  Handing seven elected politicians a searchable book of accusations against their own constituents is a political weapon, and nothing about holding the office requires it.
- **No writes at all**
  A notice goes out over the barangay's name, so posting stays with the captain and the clerk. A council seat is not a megaphone.

5 Public document check

Who uses it. Anybody holding a printout: a bank, an employer, a clerk at another counter. No account.

Scanning the QR on any document this system issues opens one page, ahead of even the session check. It answers for a municipal document, a barangay certificate and a receipt alike, and says which it found.

The check

A check that needs a login is a check that does not happen. Nothing is disclosed that the holder of the paper does not already have in their hand.

Scan or type the code
The QR carries a link, not a bare code — a phone knows what to do with a URL and nothing with sixteen loose characters.

Valid
Type, reference, who holds it, when it was issued, what it is valid until, live status.

Lapsed
Amber, and in words. A real document that needs replacing, not a fake one.

Withdrawn
Red. Adds whether a replacement exists — the difference between "do not accept this" and "do not accept this, ask for the current one".

Not found
A typo or a forgery. The codes are long and random, so the endpoint cannot be walked.

Checked N times
The only trace a photocopy leaves. Shown to the person deciding whether to accept the paper, because they are the one who can act on it.